Pythagoras

$x^2 + y^2 = r^2$

Trig:

Sin θ Cos θ Tan θ

$(1)\ \cos 2\theta = \cos^2\theta - \sin^2\theta$

$= (1 - \sin^2\theta) - \sin^2\theta$

$= 1 - 2\sin^2\theta$

Trig identities:

(2) $\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cos(\frac{\pi}{4} + 45)$

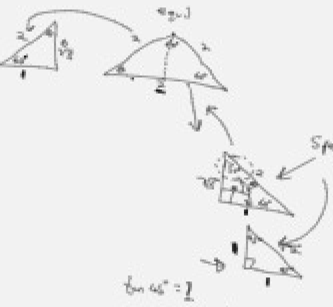
$= \cos\frac{\pi}{4} \cdot \cos 45° + \sin\frac{\pi}{4} \cdot \sin 45°$

$x^2 + y^2 = r^2$

$p^2 + y^2 = (1)^2$

$y^2 = 1 - p^2$

$y = \sqrt{1 - p^2}$ ✓



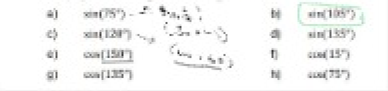
$\tan 45° = 1$

$\tan 45° = \frac{\sqrt{2}}{\sqrt{2}} = \frac{1}{1} = 1$

$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

Special angles: 30; 45°; 60°; 90;

$\sin(\alpha + \beta) = \sin\alpha \cdot \cos\beta + \cos\alpha \cdot \sin\beta$



2. Simplify the following without using a calculator:

a)	sin(75°)	b)	sin(105°)
c)	sin(120°)	d)	sin(135°)
e)	cos(150°)	f)	cos(15°)
g)	cos(135°)	h)	cos(75°)

2(a) $\sin(75°) = \sin(30 + 45)$

$= \sin 30° \cdot \cos 45° + \cos 30° \cdot \sin 45°$

$= \left(\frac{1}{2}\right) \cdot \left(\frac{1}{\sqrt{2}}\right) + \frac{\sqrt{3}}{2} \cdot \left(\frac{1}{\sqrt{2}}\right)$

$= \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$

$= \frac{(2\sqrt{2})(\sqrt{3}) + (2\sqrt{2}) \times (1)}{(2\sqrt{2})(2\sqrt{2})}$

$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$

(1) reduction

$\cos(150) = \cos(180 - 30)$

$= -\cos 30°$

$= -\frac{\sqrt{3}}{2}$ ✓

(2) reduction

$\cos(90 + 60)$

$= -\sin(60°)$

$= -\frac{\sqrt{3}}{2}$ ✓

(3) → compound d formula

$\cos(150) = \cos(90 + 60)$

$= \cos 90 \cos 60 - \sin 90° \sin 60°$

$= 0 \cdot \left(\frac{1}{2}\right) - (1) \cdot \left(\frac{\sqrt{3}}{2}\right)$

$= -\frac{\sqrt{3}}{2}$ ✓

Addition of fractions

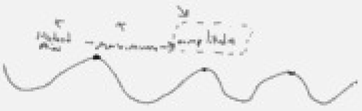
$\frac{1}{2} + \frac{2}{3}$

$\frac{1 \times 3 + 2 \times 2}{2 \times 3}$

$= \frac{3 + 4}{6}$

$= \frac{7}{6}$

Trig functions

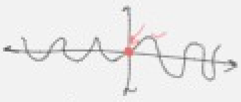


circle:

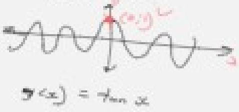
Domain (x-values)

Range (y-values)

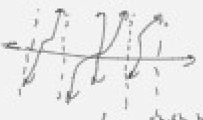
$f(x) = \sin x$



$h(x) = \cos x$



$g(x) = \tan x$



$f(x) = a \sin b(x - p) + q$

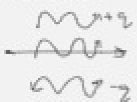
shift to the left and right

↑ shift up

↓ shift down

amplitude

Period $= \frac{360}{b}$



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